Network Science

Lecture 01: Introduction

Prof. Dr. Michael Granitzer

Dr. Kanishka Ghosh Dastidar

Dr. Jelena Mitrovic

Welcome to Network Science

> Modelling, analysing and infering digitial and real-world phenomena through networks.

Today

- what is a network?
- why do networks matter?
- what can we analyse with networks?
- what will you learn in this course?

Networks

Guiding Question: What do you think a network is?

Your Everyday Networks

Think about networks you encounter every day.

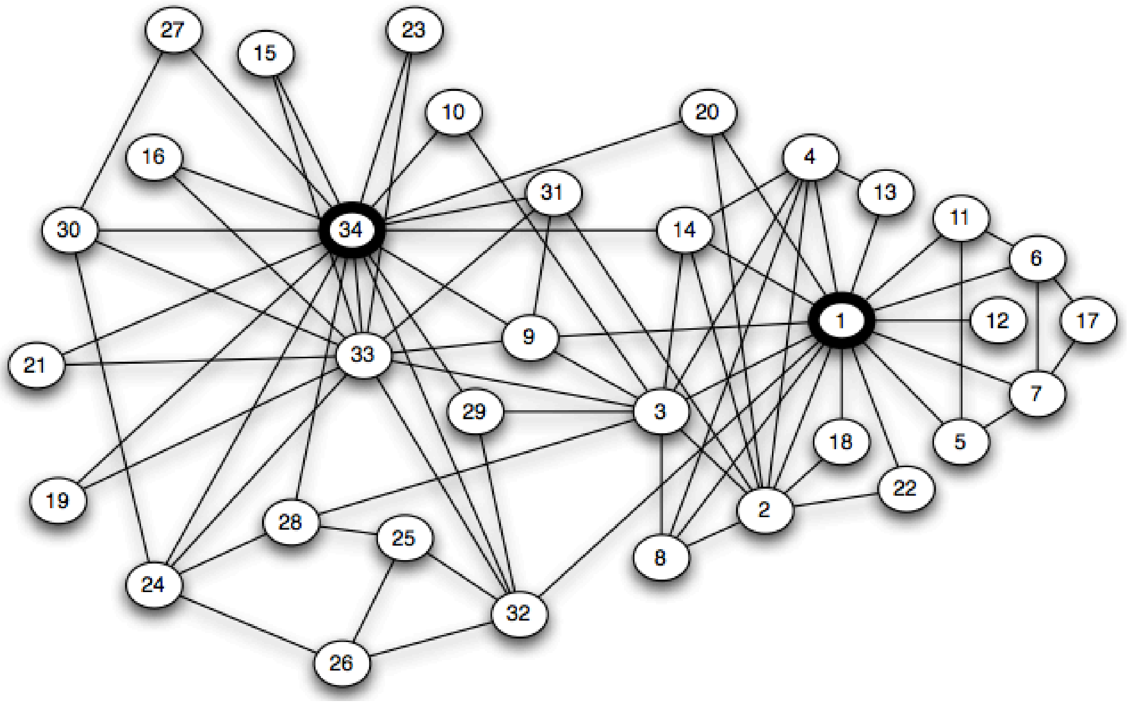
Your Everyday Networks

Think about networks you encounter every day.

- Who did you talk to today? → **social network**
- Which websites link to each other? → **information network**
- Which companies trade with each other? → **economic network**
- Which roads connect which cities? → **infrastructure network**

Social Networks

Link structures can reveal friendships, communities, and influence.

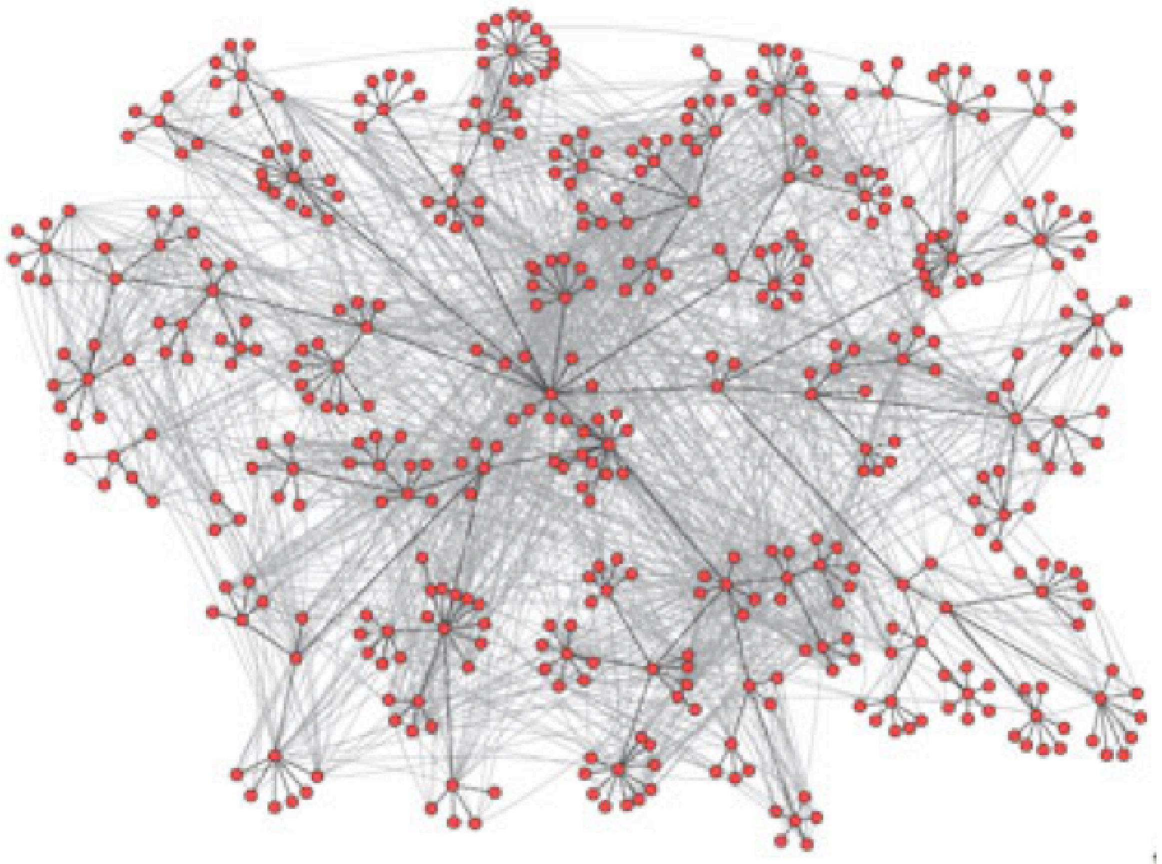


Source: Zachary 1977 — modified; Original data: Zachary 1977; visualization redrawn

Zachary's Karate Club (1977). Each node is a club member, each edge a friendship. The club split into two factions — visible as two dense clusters. One of the most-studied small social networks.

Communication Networks

Link structures can reveal information flow and potential dissemination routes.

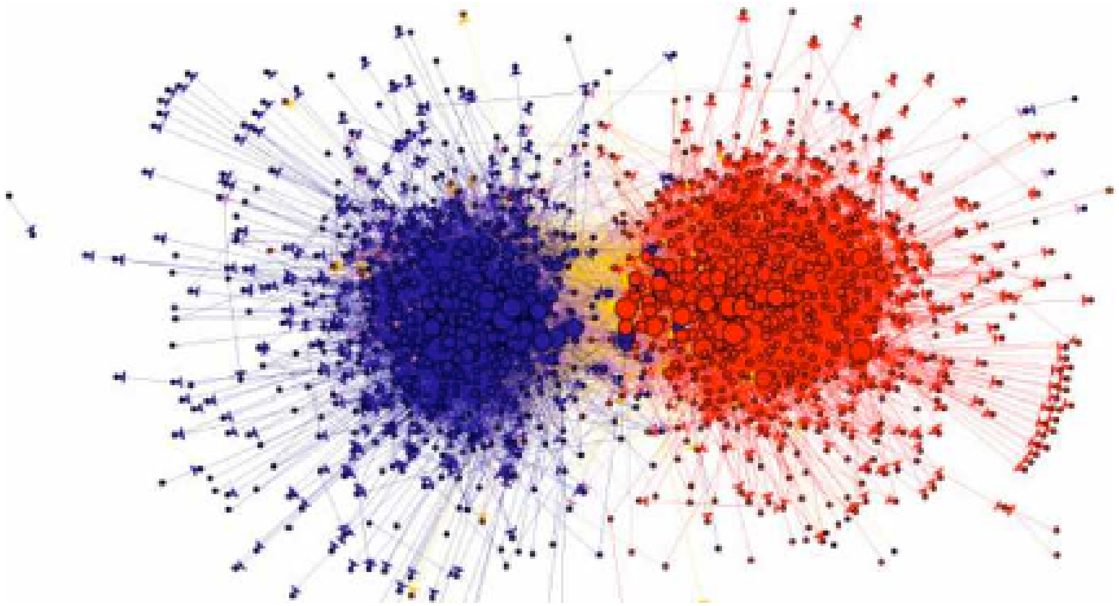


Source: Easley & Kleinberg 2010, p. Ch. 2

Email communication in an organization. Nodes represent employees, edges represent email exchanges. The actual communication topology differs substantially from the formal hierarchy.

Information Networks

Link structures can reveal clusters, polarization, and community structure — before reading a single word.

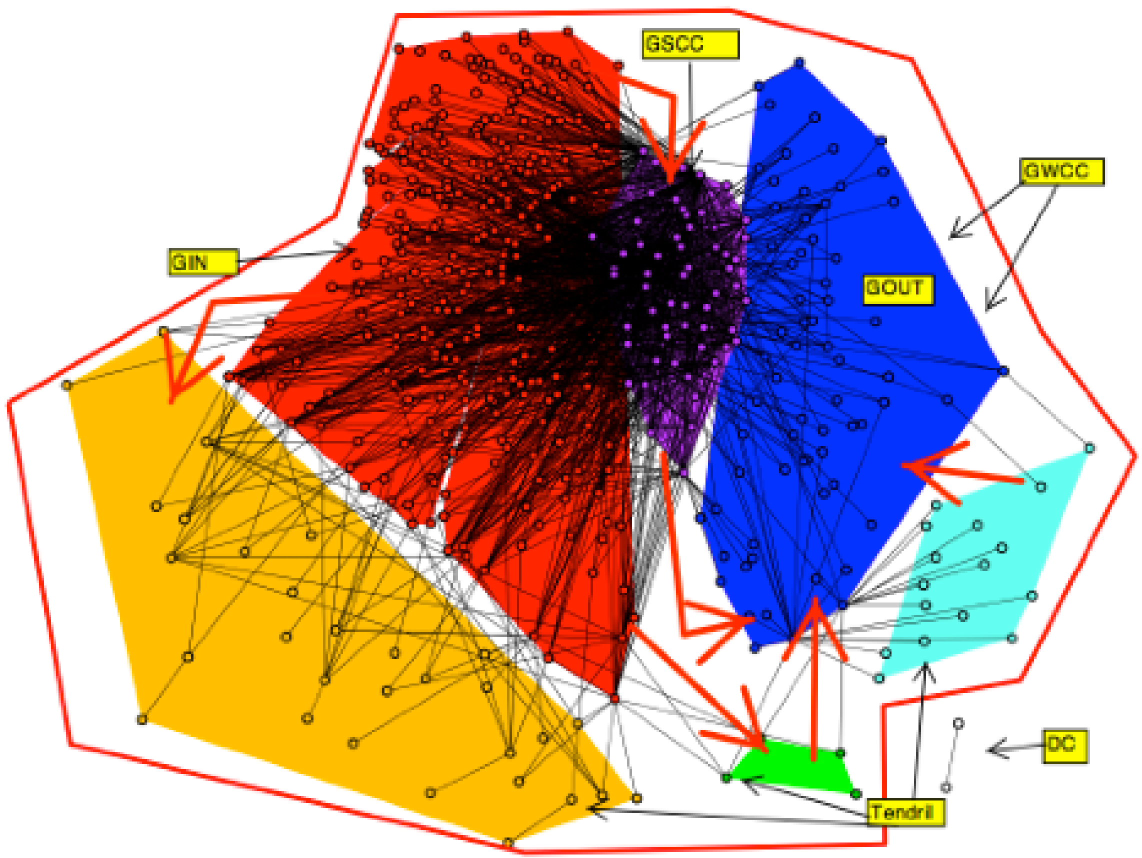


Source: Adamic & Glance 2005

Political blog network (2004 US election). Nodes are political blogs, edges are hyperlinks between them. The two dense clusters correspond to liberal and conservative blogs — structure reveals political alignment without reading any content.

Economic Networks

Financial systems form networks of dependencies and influence (Easley & Kleinberg, 2010).



Source: Easley & Kleinberg 2010, p. Ch. 1

Interbank loan network. Nodes are financial institutions, edges represent lending relationships. When one institution defaults, the impact cascades through the network — structure determines systemic risk.

What Do All These Examples Have in Common?

What Do All These Examples Have in Common?

- There are **things** (people, pages, institutions, cities).
- There are **connections** between them (friendships, links, loans, roads).
- The pattern of connections creates **structure** that matters.

What Do All These Examples Have in Common?

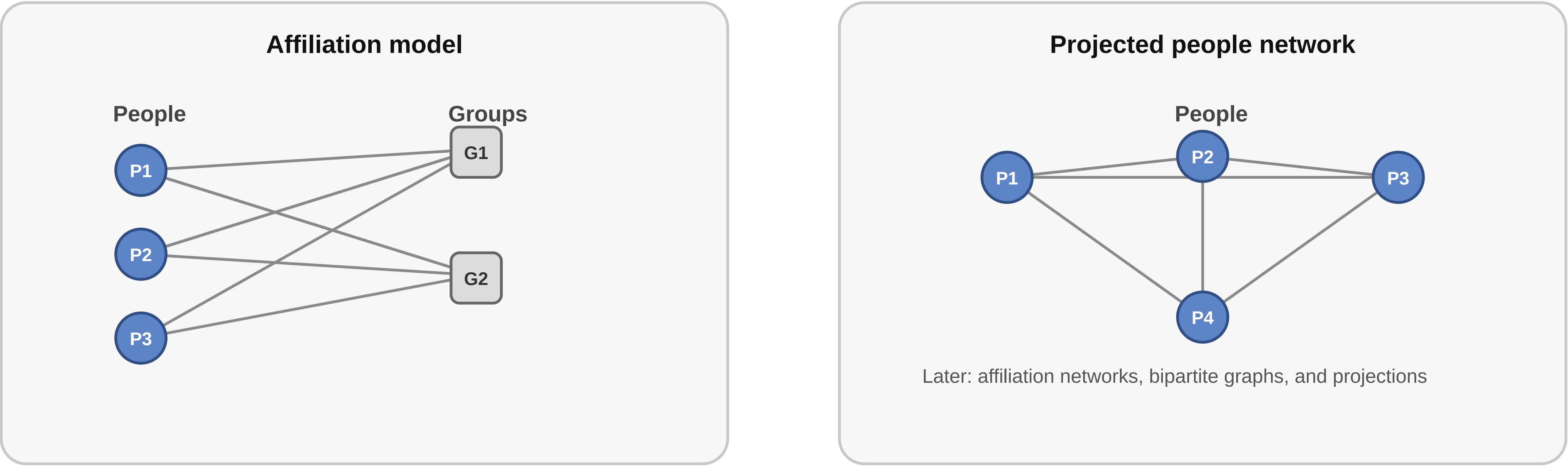
- There are **things** (people, pages, institutions, cities).
- There are **connections** between them (friendships, links, loans, roads).
- The pattern of connections creates **structure** that matters.

$$\text{Network} = \text{Entities} + \text{Relationships between them}$$

What Gets Lost in a Network Model?

When we turn a real situation into a network, we always leave something out.

Same situation, different network models



> The same relational setting can be modeled in more than one way. An affiliation structure can become a bipartite network or a projected one-mode graph depending on the question.

Takeaway

A network is a structured collection of entities and the relationships between them. Understanding networks means understanding how to manage, describe, and reason about these entities and their connections.

Quick Check

Which of the following statements about networks are correct?

- A network consists of entities (nodes) and relationships between them (edges)
- The same network abstraction applies across very different domains
- Networks always have directed edges
- A network model captures every detail of the underlying system
- Edges can carry additional information such as weight or type

Quick Check — Solution

Which of the following statements about networks are correct?

- **Correct:** A network consists of entities (nodes) and relationships between them (edges)
- **Correct:** The same network abstraction applies across very different domains
- **Incorrect:** Networks always have directed edges (edges can also be undirected)
- **Incorrect:** A network model captures every detail of the underlying system (they are models and have lossy abstractions)
- **Correct:** Edges can carry additional information such as weight or type

Impacts of Networks

> **Guiding Question:** Why do networks matter — what can we learn from connections?

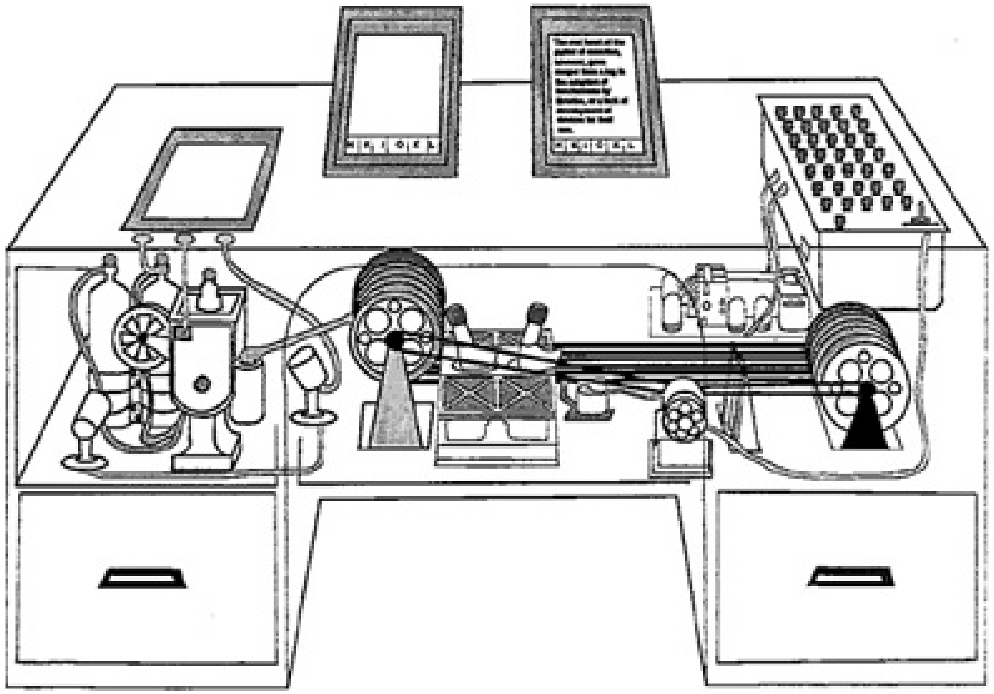
The Whole Is More Than Its Parts

- Knowing every person in a city does not tell you how information travels — you need to know **who talks to whom**.
- Knowing every website does not tell you which ones are easy to find — you need to know **which pages link to which.**
- Knowing every bank does not tell you which failure would cause a cascade — you need to know **who has financial exposure to whom.**

$$\text{Outcome} \neq \text{actor properties alone}$$

Practical Example: Linked Information

Vannevar Bush's MEMEX anticipated linked information systems and associative navigation (Bush, 1945).

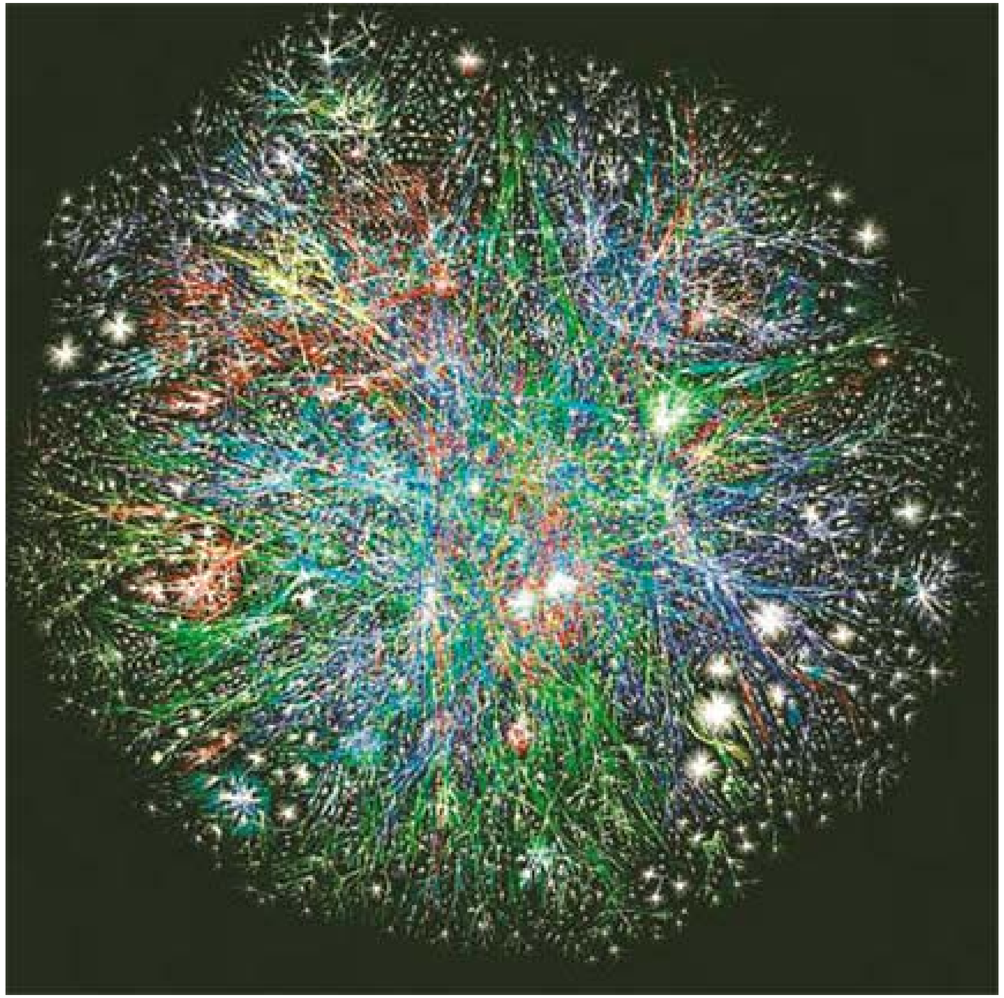


Source: Bush 1945 — public-domain

Vannevar Bush's MEMEX (1945). A conceptual device for storing and linking documents through associative trails. It anticipated the idea that information becomes valuable through connections — the intellectual ancestor of hypertext and the Web.

The Web as a Network

The Web operationalized hypertext at planetary scale: linked documents, user navigation, and a growing information network (Berners-Lee, 1990).

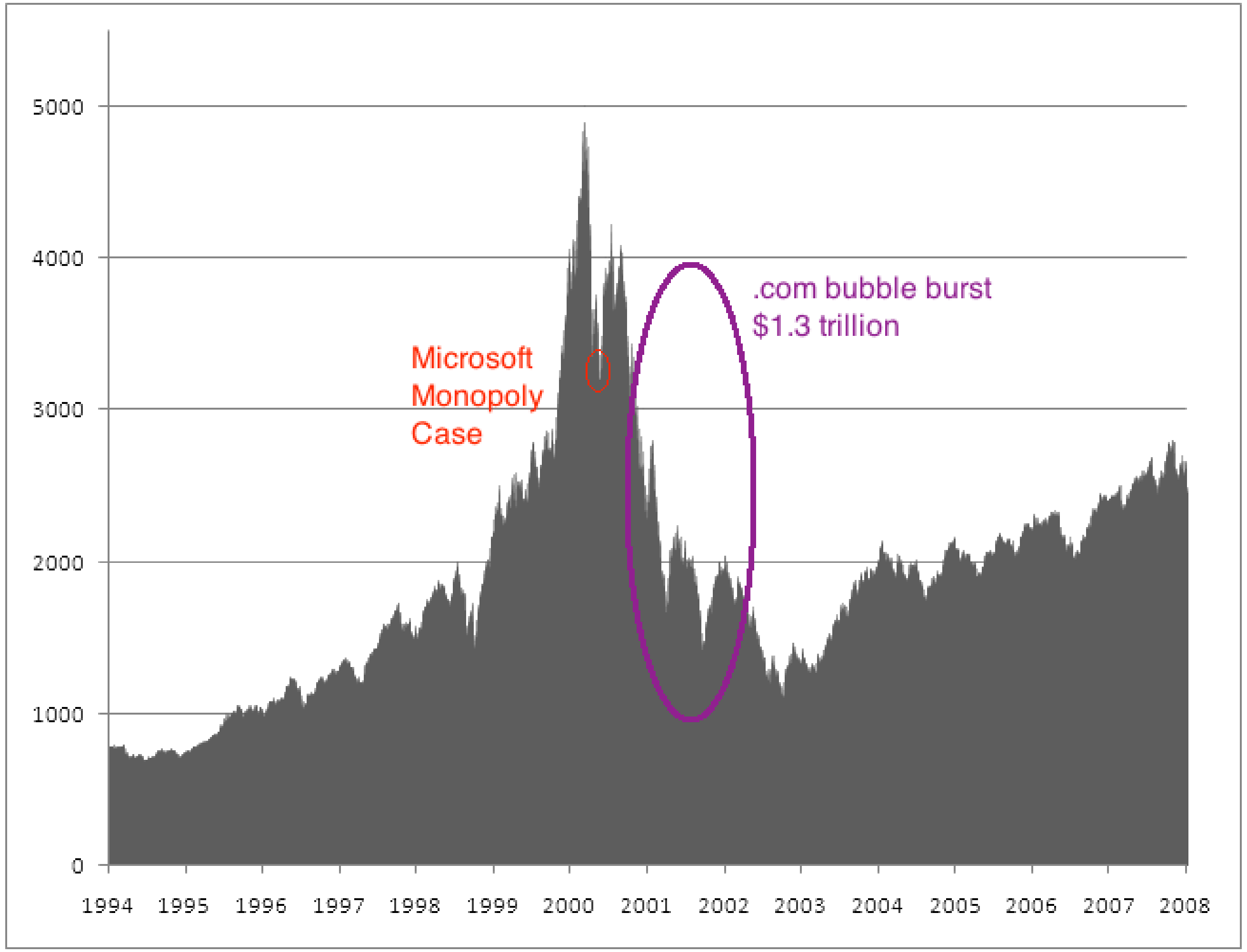


Source: Easley & Kleinberg 2010

Visualization of internet routing paths. Each node represents a router or autonomous system, edges are data connections. The dense core and sparse periphery illustrate the uneven topology of real technical networks.

Practical Example: Platform Power

Surviving web companies learned to exploit network effects — users became part of the value-creation process (O'Reilly, 2005).



Source: Public market data, NASDAQ composite index

NASDAQ composite index around the dot-com bubble (1995–2005).
The sharp rise and crash illustrate how many early web companies failed. The survivors — Google, Amazon, Facebook — were those that understood and exploited network effects.

What Can We Learn from Networks?

- **Who is important?** → centrality, influence
- **Are there groups?** → communities, clusters
- **How do things spread?** → diffusion, contagion
- **How robust is the system?** → fragility, resilience
- **What shapes attention?** → visibility, ranking, filtering

Takeaway

Networks matter because structure shapes visibility, coordination, influence, and lock-in. Looking at connections reveals what looking at individuals alone cannot.

Quick Check

Why are network models useful for studying platforms like the Web?

A. They allow us to ignore structure and focus on individual users.

B. They reveal how connections create effects like lock-in and influence.

C. They primarily help in analyzing the content of individual web pages.

D. They show that all users have equal influence in a system.

Quick Check — Solution

Why are network models useful for studying platforms like the Web?

Correct: B. They reveal how connections create effects like lock-in and influence.

The Web's power comes not just from the individual pages (nodes), but from how they are linked (edges), which drives visibility and platform power.

Network Science & Network Analysis

> **Guiding Question:** What phenomena can we actually analyse with networks?

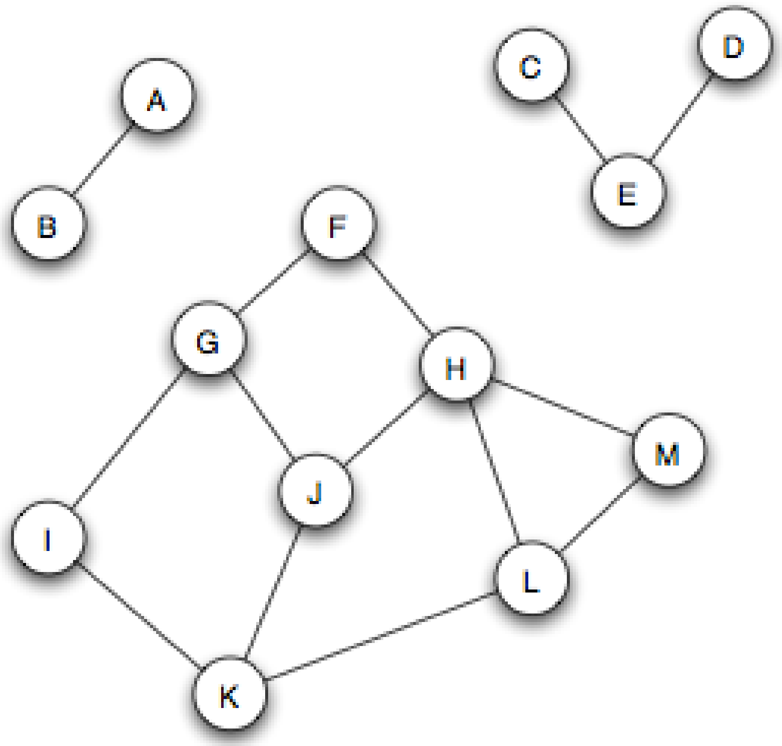
From Intuition to Analysis

We already said networks reveal structure. But what kinds of questions can we actually answer?

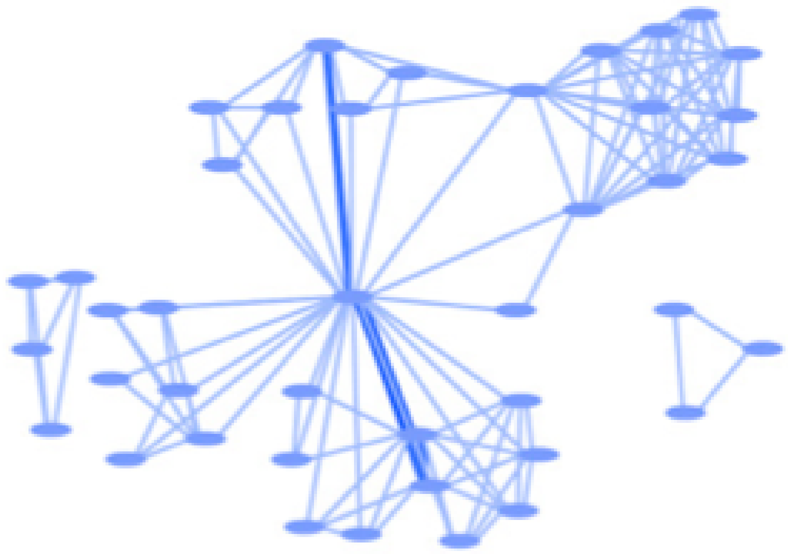
- **Structure**: How is the network organized? Are there clusters, hubs, or bottlenecks?
- **Position**: Who sits in a central or peripheral position? Who bridges groups?
- **Dynamics**: How does information, disease, or opinion spread through the network?
- **Evolution**: How does the network change over time? What drives growth?

Analysing Structure: Communities

Networks often have dense internal clusters that correspond to real groups.



Source: Easley & Kleinberg 2010



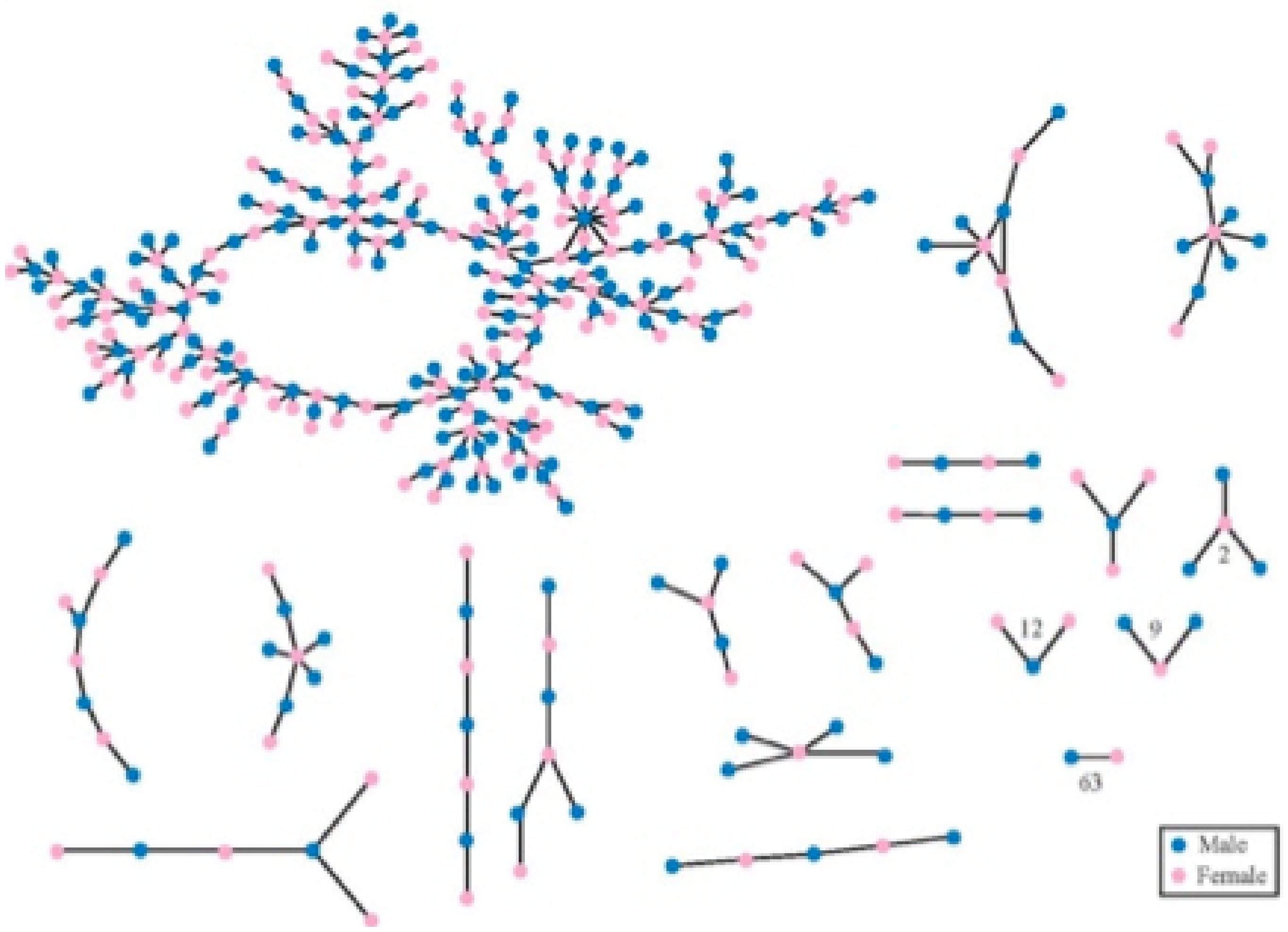
Source: Easley & Kleinberg 2010

Three separate connected components — groups of nodes with no links between them.

A scientific collaboration network where clusters represent research groups.

Analysing Dynamics: Diffusion and Spreading

How a rumor, virus, or innovation spreads depends critically on network structure.



Source: Easley & Kleinberg 2010

High school friendship network.
Nodes are students, edges are reported friendships. The structure shows a dense core and peripheral chains — a rumor starting in the core spreads far faster than one starting at the edge.

Core Questions of Network Science

Across all these domains, network science asks a common set of structural questions:

- How do we **describe** a network formally?
- Which structural **properties** matter?
- How do networks shape **diffusion** and coordination?
- Which common **patterns** appear across different domains?

Takeaway

> Network science gives us concrete tools to analyse structure, position, dynamics, and evolution. The same analytical questions apply whether we study friendships, web links, financial flows, or disease transmission.

Quick Check

Match the network phenomenon with the type of analysis used to study it:

1. Finding groups of friends in a high school
2. Measuring how fast a rumor spreads
3. Identifying the most influential blogger

A. Dynamics (Diffusion) B. Structure (Communities) C. Position (Centrality)

Quick Check — Solution

Match the network phenomenon with the type of analysis used to study it:

- **1 - B. Structure (Communities)**
- **2 - A. Dynamics (Diffusion)**
- **3 - C. Position (Centrality)**

Curriculum

> **Guiding Question:** What do you want to learn about networks?

What You Will Learn

Course Topics

1. **Graph theory** — the formal language for describing networks
2. **Social networks** — ties, communities, and information flow
3. **Centrality and influence** — who matters and why
4. **Community detection** — finding groups in networks
5. **Information networks and the Web** — links, search, and ranking
6. **Network dynamics** — diffusion, contagion, and cascades

Course Goal

Use network analysis to understand and design social information systems.

- not just learn definitions — **interpret** formal concepts
- not just run algorithms — **critique** the models behind them
- not just describe networks — **reason** about their consequences

Recommended Reading

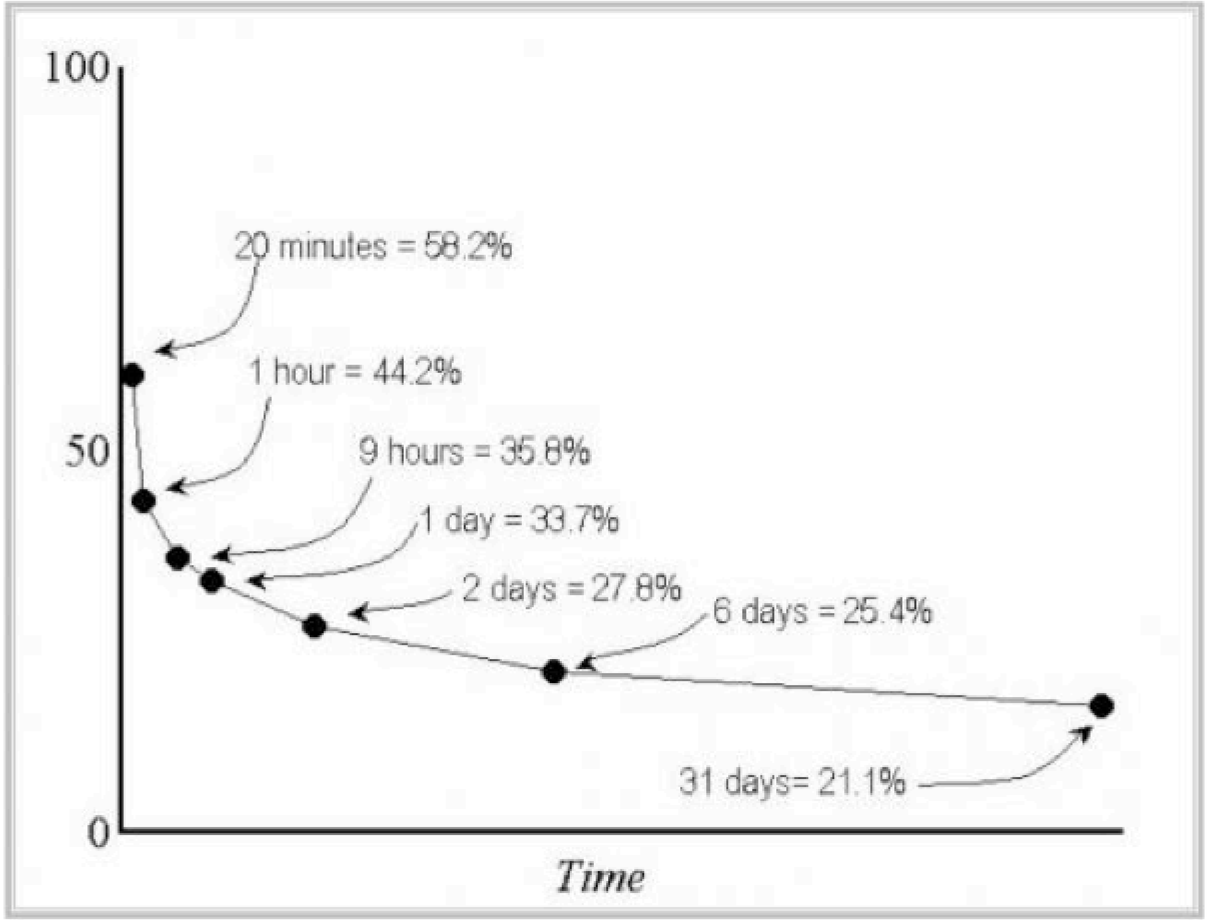
- David Easley and Jon Kleinberg, *Networks, Crowds, and Markets* (Easley & Kleinberg, 2010)
- Mark Newman, *Networks: An Introduction* (Newman, 2010)

How to Learn This Course

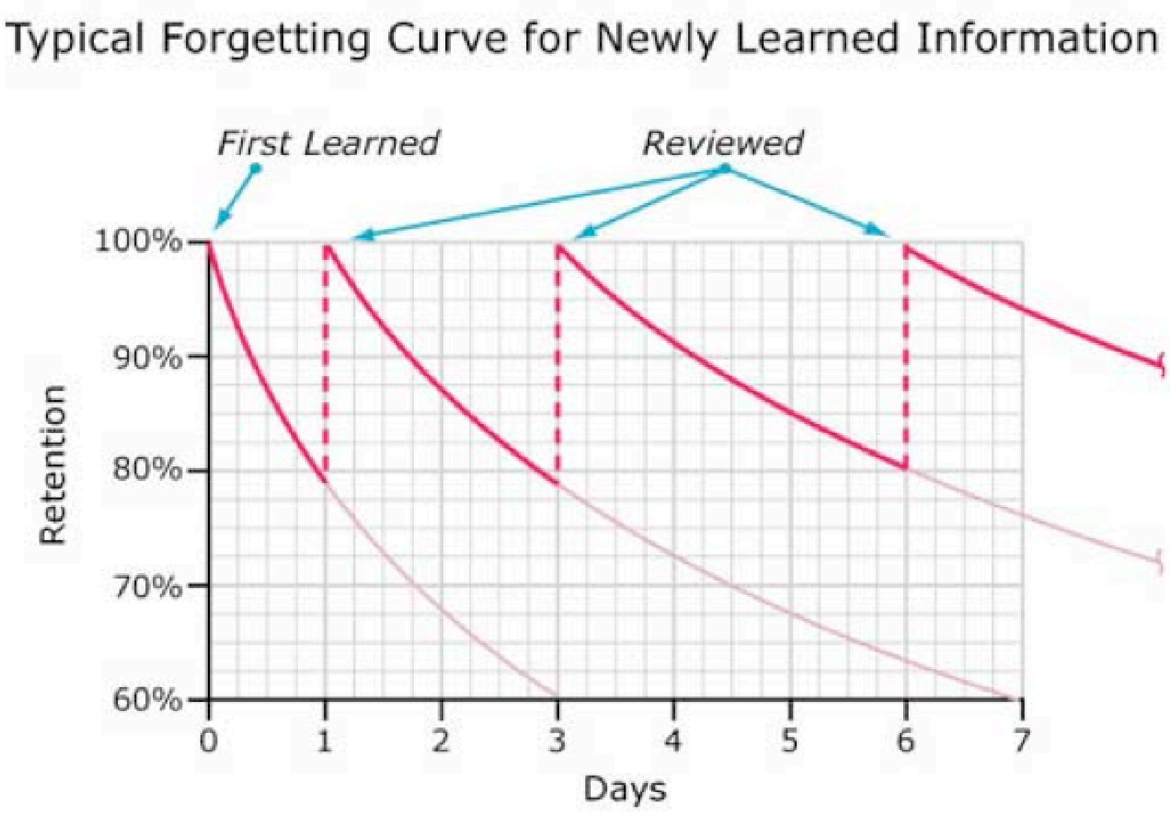
- **preview** slides before class
- **discuss** key ideas in small groups
- **work through** exercises yourself before comparing solutions
- **use AI tools** to accelerate access to explanations — but if you cannot explain a concept without the tool, you do not yet own it

Forgetting and Repetition

Repeated contact with the same concept reduces forgetting and improves retention.



Source: Ebbinghaus forgetting curve, redrawn



Source: Ebbinghaus curve with spaced repetition, redrawn

Takeaway

> The course is about modeling, interpretation, and critique — not only about definitions. You will build a toolkit for understanding networks that applies across social, technical, and economic systems.

Image Credits & References

Adamic, Lada A. and Glance, Natalie (2005). The political blogosphere and the 2004 U.S. election: divided they blog. *Proceedings of the 3rd International Workshop on Link Discovery (LinkKDD '05)*.

Bush, Vannevar (1945). As We May Think. *The Atlantic Monthly*.

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning about a Highly Connected World. *Cambridge University Press*. https://www.cs.cornell.edu/home/kleinber/networks-book/

Zachary, Wayne W. (1977). An Information Flow Model for Conflict and Fission in Small Groups. *Journal of Anthropological Research*.